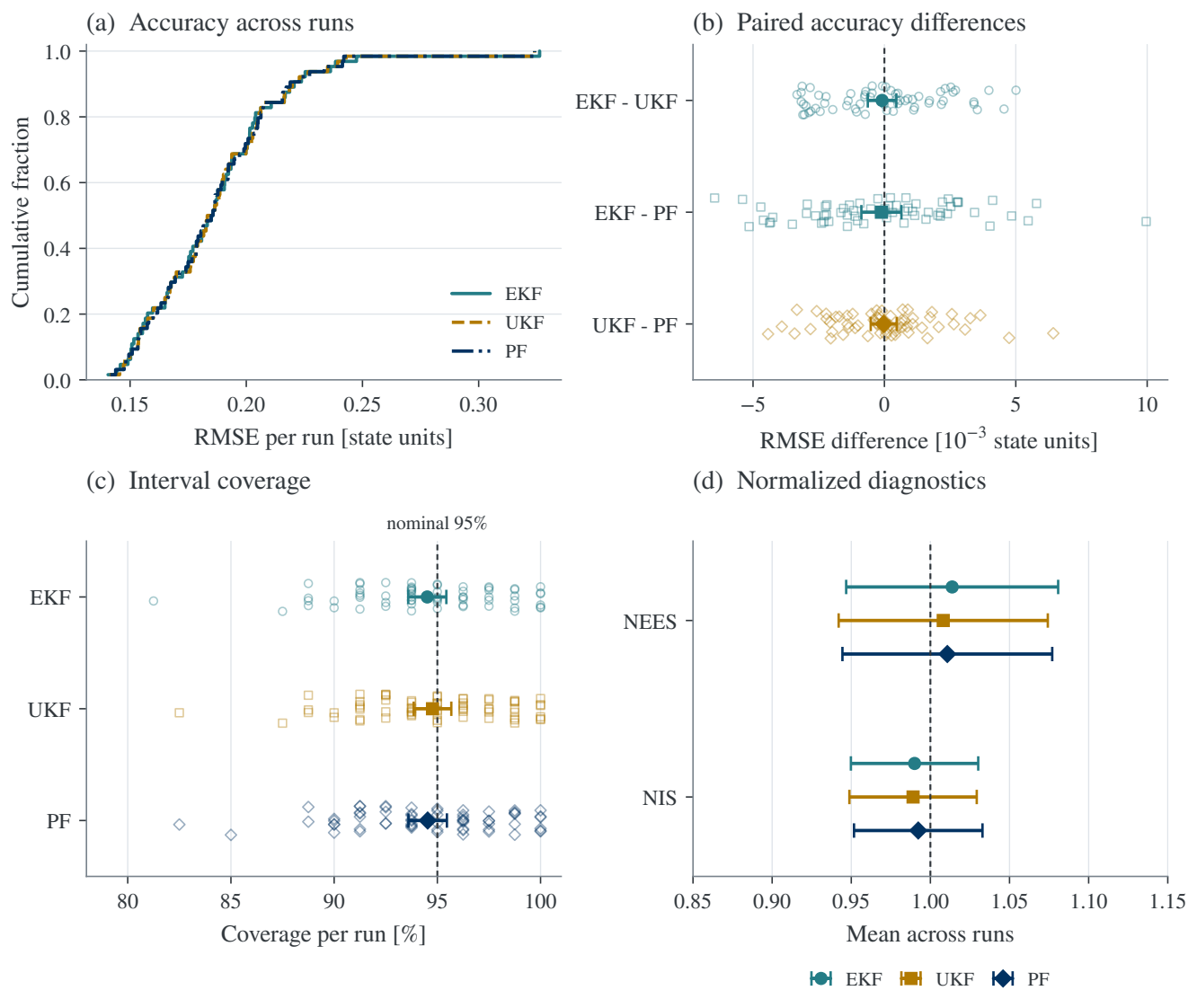
(a) Accuracy across runs
Cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
0.15
0.20
0.25
0.30
RMSE per run [state units]
EKF
UKF
PF
(b) Paired accuracy differences
EKF - UKF
EKF - PF
UKF - PF
−5
0
5
10
RMSE difference [10−3 state units]
(c) Interval coverage
nominal 95%
EKF
UKF
PF
80
85
90
95
100
Coverage per run [%]
(d) Normalized diagnostics
NEES
NIS
0.85
0.90
0.95
1.00
1.05
1.10
1.15
Mean across runs
EKF
UKF
PF